

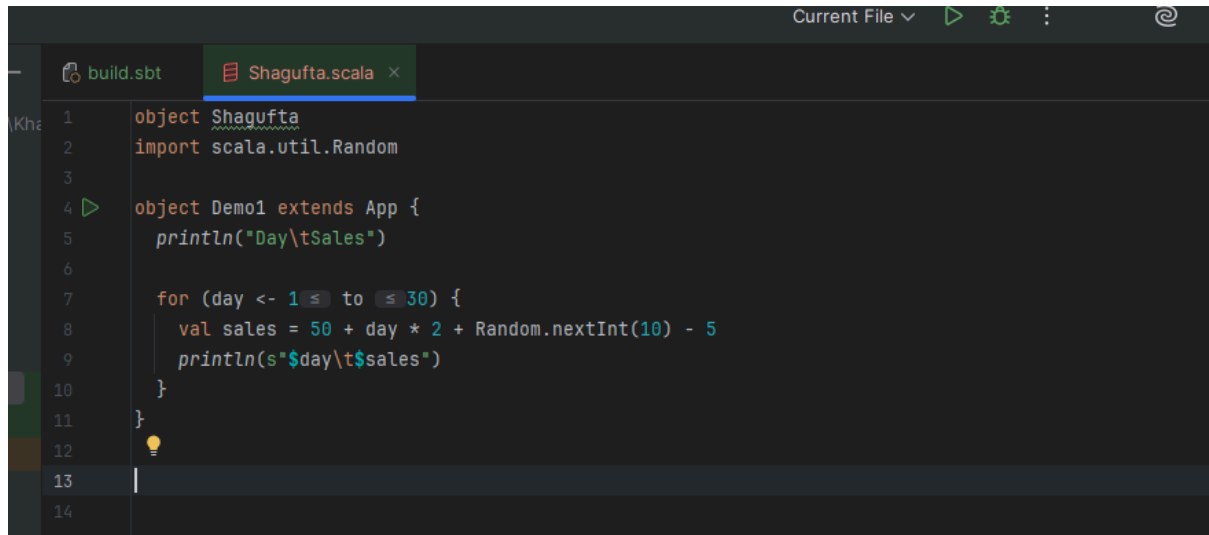
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Scala Practical - 14

Roll No. :S018	Name:Khan Shagufta
Class :SYIT	Batch : 1
Date of Assignment :11/08/26	Date/Time of Submission :11/08/26

1)Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month)

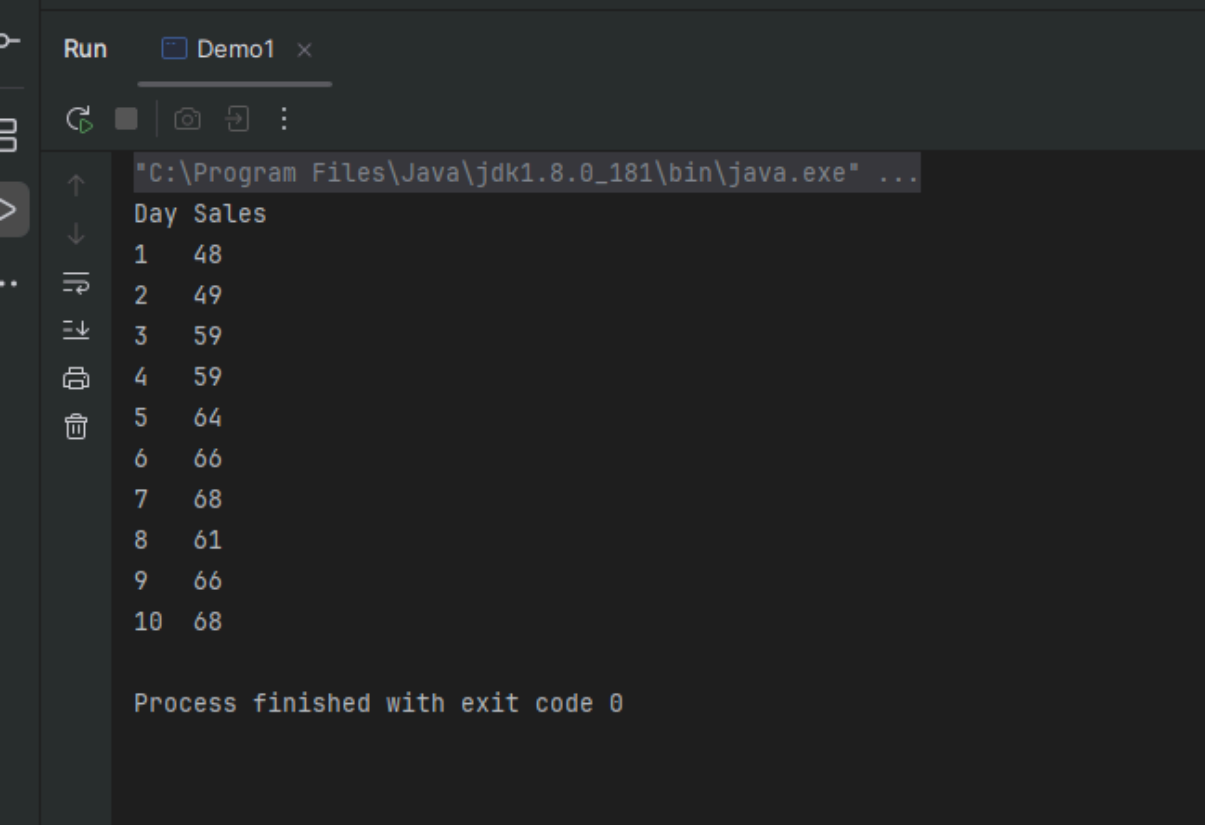
Code:

A screenshot of a code editor with a dark theme. The editor has two tabs at the top: 'build.sbt' and 'Shagufta.scala'. The 'Shagufta.scala' tab is active. The code is written in Scala and is as follows:

```
1 object Shagufta
2 import scala.util.Random
3
4 object Demo1 extends App {
5     println("Day\tSales")
6
7     for (day <- 1 to 30) {
8         val sales = 50 + day * 2 + Random.nextInt(10) - 5
9         println(s"$day\t$sales")
10    }
11 }
12
13
14
```

The code generates 30 lines of synthetic data, each representing a day and its sales. The sales value is calculated as 50 plus twice the day number, plus a random integer between 0 and 9, minus 5. The output is printed in a tab-separated format.

Output:



```
Run Demo1 x
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Day Sales
1 48
2 49
3 59
4 59
5 64
6 66
7 68
8 61
9 66
10 68

Process finished with exit code 0
```